

Speaker and Language Recognition – From Laboratory Technologies to the Wild

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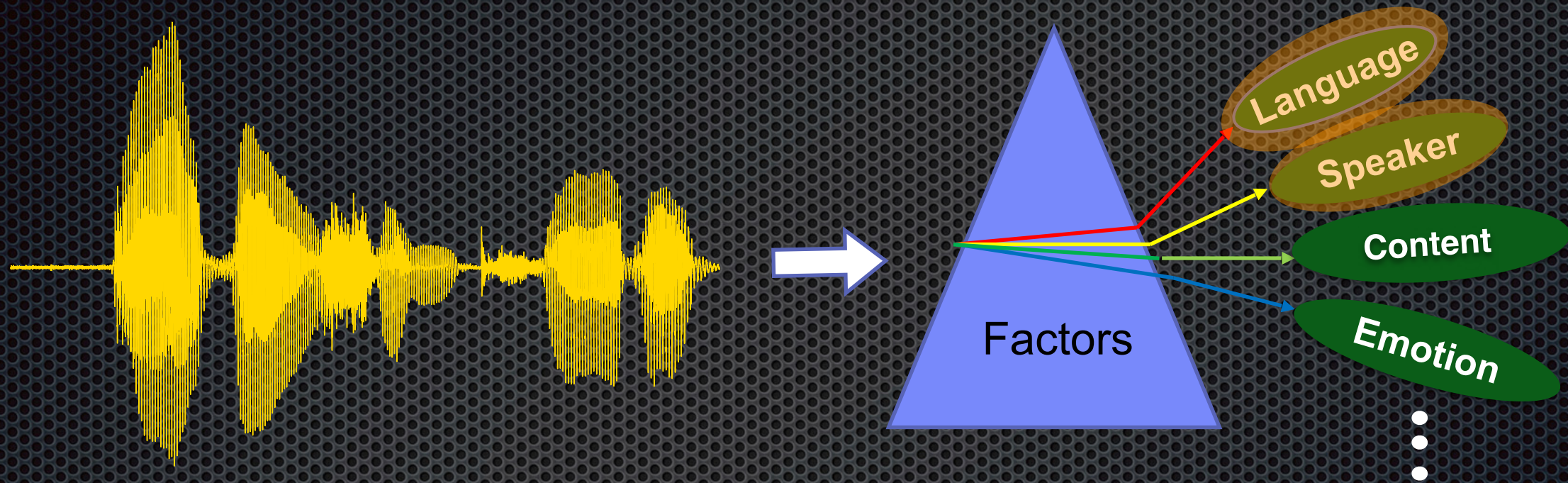


Roadmap

- ✦ Problem Definition
- ✦ Challenges Involved
- ✦ Historical Perspective
- ✦ Breakthroughs
- ✦ Open Issues and Future Research

Problem Definition

- Speech - multitude of intrinsic information sources.



- Extrinsic sources - noise, channel, reverberation.
- Question - can we separate the sources ?

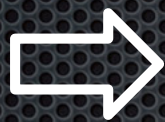
Problem Definition

Speaker Identification

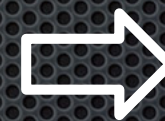
Identify among a set of speakers ?



Speaker Verification



Reject !

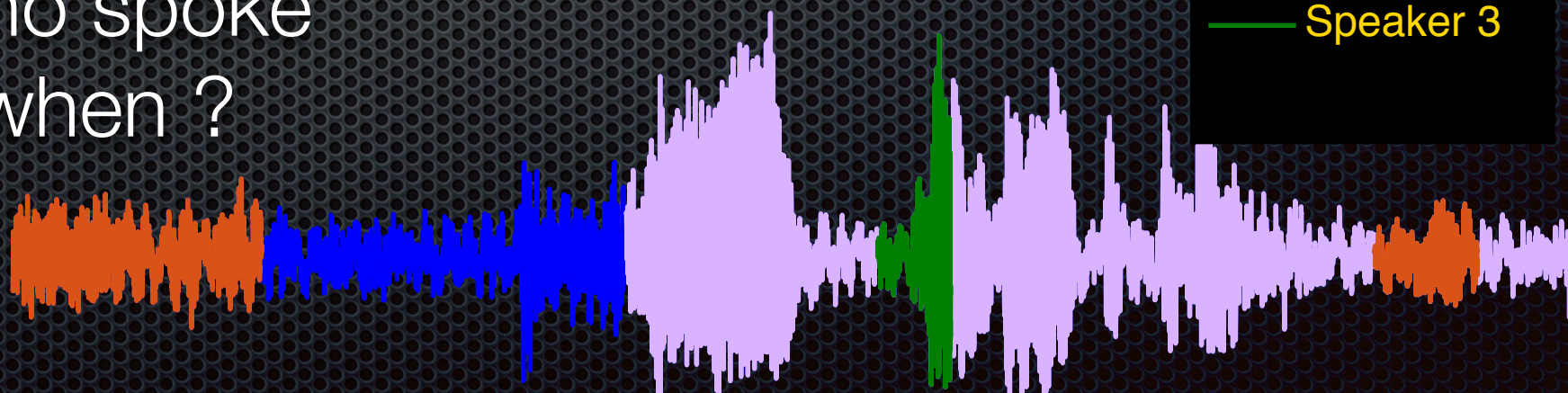


ACCEPT

Speaker Diarization

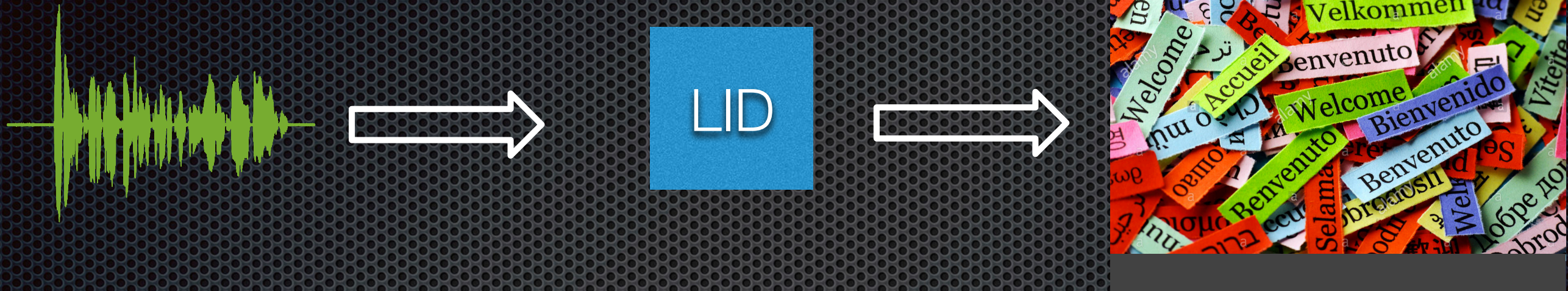
Who spoke when ?

— Speaker 0
— Speaker 1
— Speaker 2
— Speaker 3



Problem Definition

Language Identification

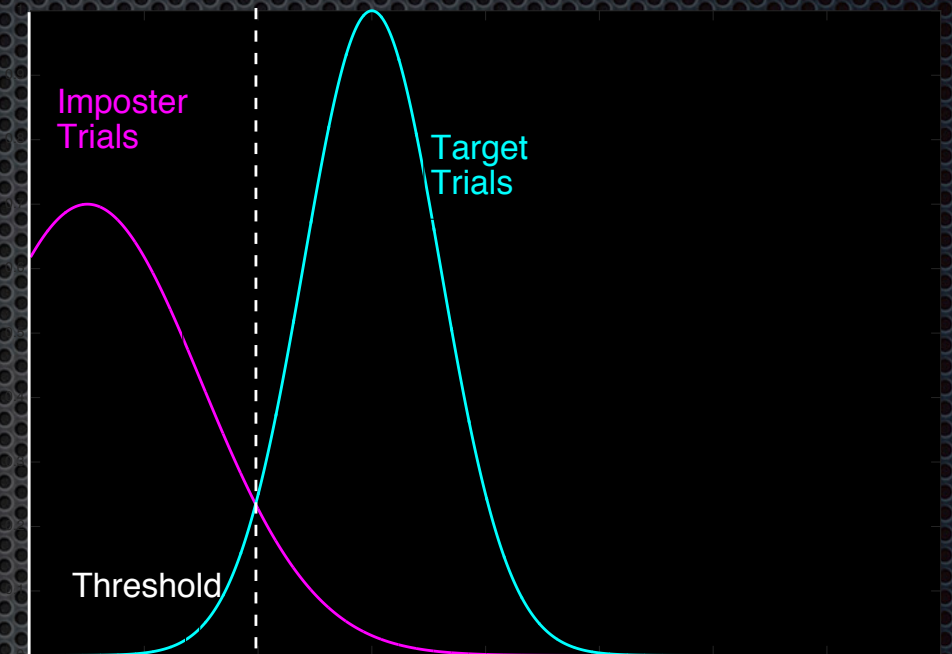


Language Verification



Performance Measures (Verification)

- Miss (false negative) rate versus false alarm (false positive) rate - Detection Error Tradeoff (DET) curve.



- Equal Error Rate (EER)

$$\theta^* = \theta : P_{miss}(\theta) = P_{fa}(\theta) \quad EER = P_{miss}(\theta^*)$$

- Min Detection Cost Function (DCF)

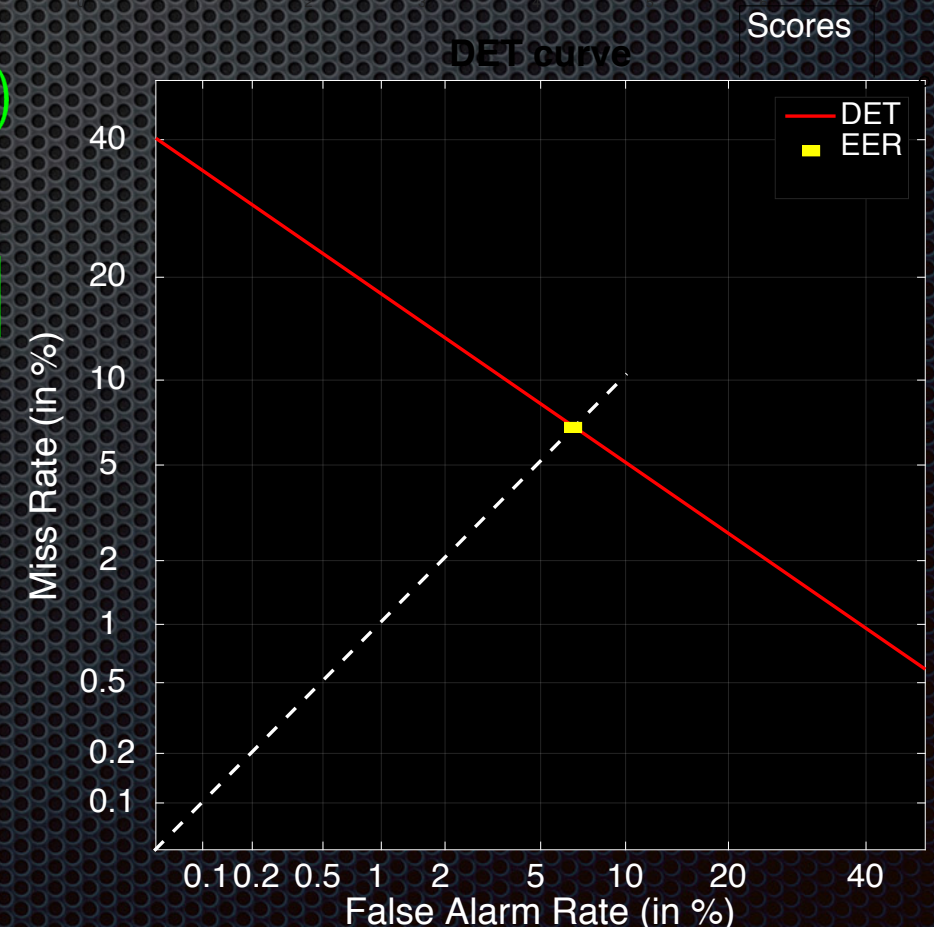
$$\min DCF = \min_{\theta} [0.1P_{miss}(\theta) + 0.99P_{fa}(\theta)]$$

- Normalized cost function

$$C_{norm}(\theta) = P_{miss}(\theta) + \beta P_{fa}(\theta)$$

- Average cost function for $\theta = \log(\beta)$

$$C_{avg} = 0.5(C_{norm}^{\beta_1} + C_{norm}^{\beta_2})$$



Applications

■ Commercial

- Multi-lingual speech processing.
- Biometric authentication.
- Customized wake-word detection.
- Speaker diarization for speech recognition.
- Profiling and speaker characterization.

■ Forensics and surveillance

Challenges

- Environmental distortion
 - In telephone or far-field microphone conditions
 - Presence of noise sources
- Duration constraints
 - Moving to short duration (from minutes to few seconds).
- Interference from other speech cues
 - Influence of spoken content
 - Other paralinguistic factors related to emotion, accent etc.



Scope of this talk

- Unified view of the evolution of speaker verification and language identification research.
 - Biased sampling of the broad literature and review of various methods
- Illustrations with NIST Speaker and Language Recognition Evaluation Challenges
- More Reading,
 - ✓ “Spoken Language Recognition : From fundamentals to practice”, Li et al, IEEE SP Magazine, 2013
 - ✓ “An overview of text-independent speaker recognition: From features to supervectors”, Kinnunen et al, Speech Comm. 2010.



Roadmap

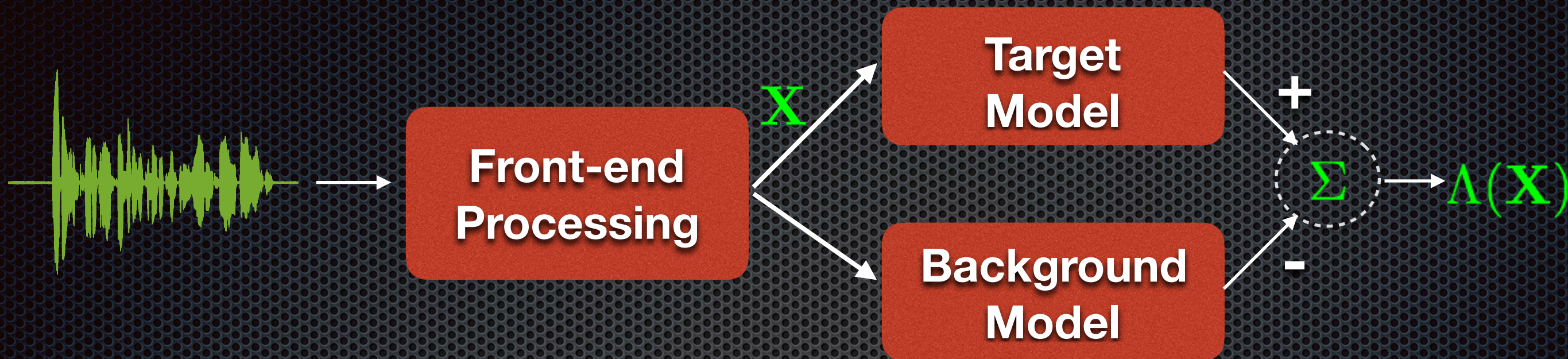
- ✦ Problem Definition
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- ✦ Historical Perspective and Breakthroughs
- ✦ Open Issues and Future Research

Historical Perspective < 2000

- The early combustion engine model



Background Model Based Approach



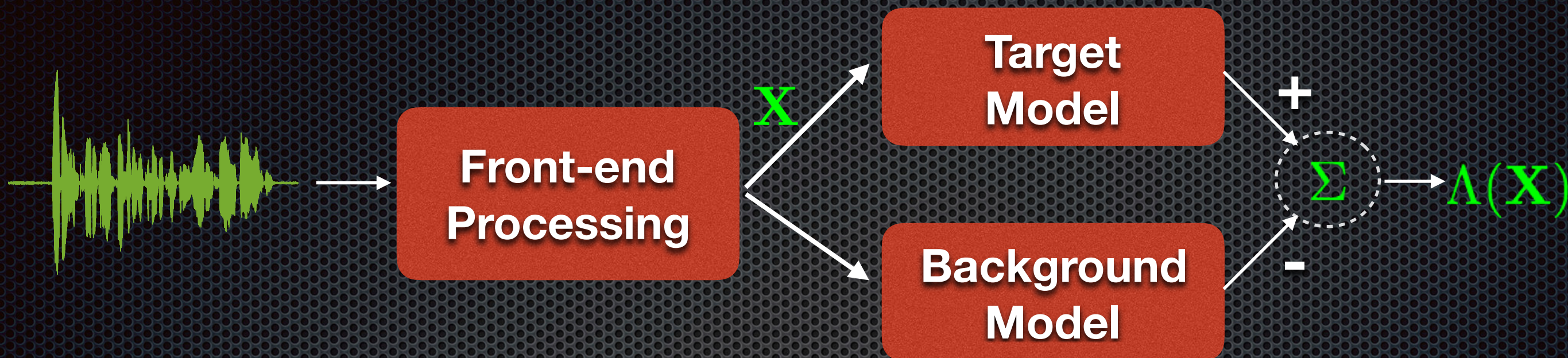
- Front-end processing

- converting audio signal to features

- ✓ most common features are the mel frequency cepstral coefficients (MFCC) with delta and delta-delta components.



Background Model Based Approach



- Background model - Gaussian mixture model

$$p(\mathbf{x}|\lambda) = \sum_{i=1}^M w_i p_i(\mathbf{x})$$

$$p_i(\mathbf{x}|\theta) = \frac{1}{\sqrt{(2\pi)^D |\Sigma_i|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \mu_i)^* \Sigma^{-1} (\mathbf{x} - \mu_i) \right\}$$

- Trained on data from all the speakers/languages.

Background Model Based Approach

- Target Model (adaptation to $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_T\}$)

$$p(i|\mathbf{x}_t) = \frac{w_i p_i(\mathbf{x}_t)}{\sum_{i=1}^M w_i p_i(\mathbf{x}_t)}$$

$$n_i = \sum_{t=1}^T p(i|\mathbf{x}_t)$$

Zeroth order
Stats

$$E_i(\mathbf{x}) = \frac{1}{n_i} \sum_{t=1}^T p(i|\mathbf{x}_t) \mathbf{x}_t$$

First order
Stats

$$\hat{\mu}_i = [\alpha_i E_i(\mathbf{x}) + (1 - \alpha_i) \mu_i]$$

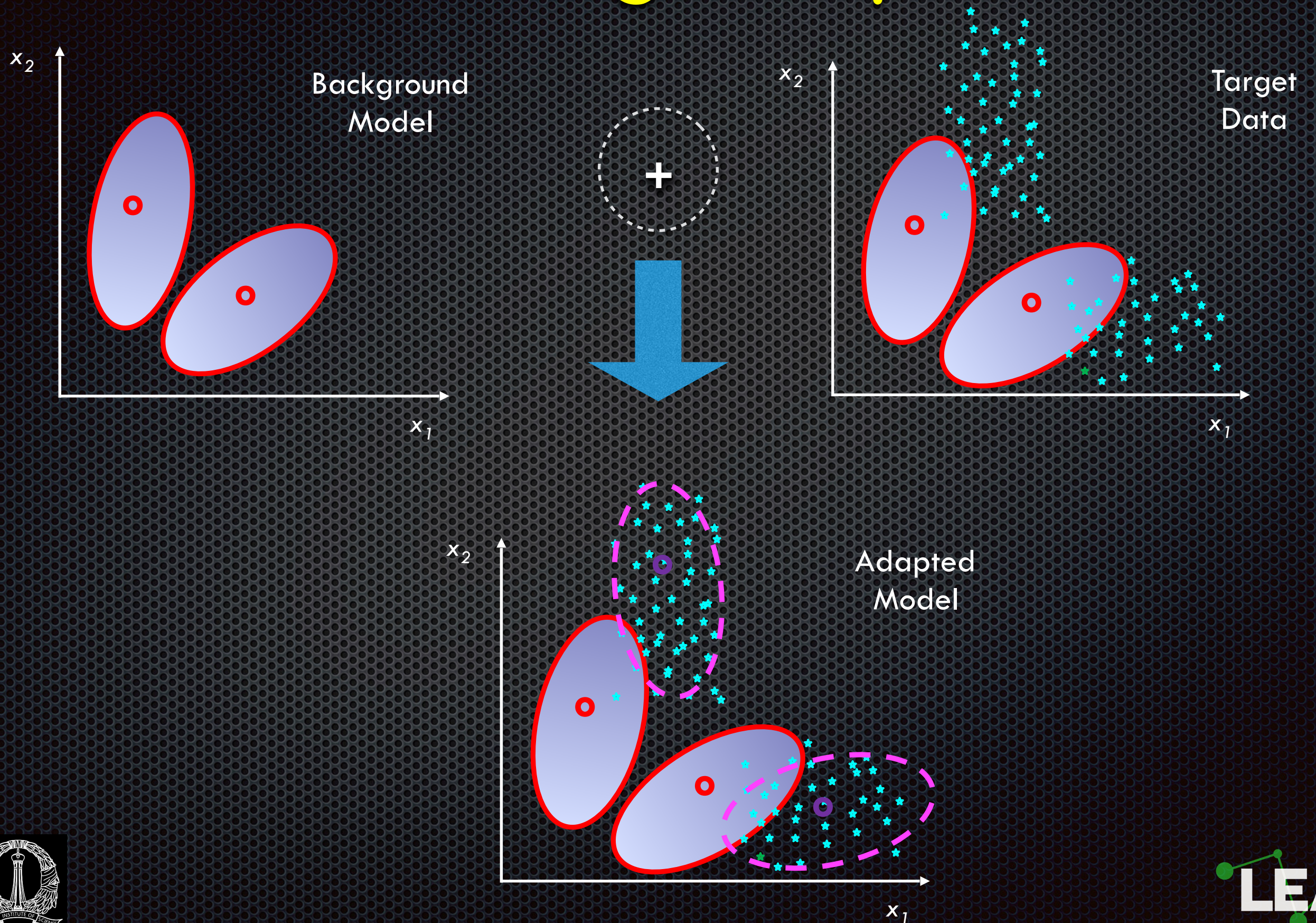
$$\alpha_i = \frac{n_i}{n_i + r}$$

- Hypothesis testing

$$\Lambda(\mathbf{X}) = \log(\mathbf{X}|\lambda_{spk}) - \log(\mathbf{X}|\lambda_{bkgrnd})$$



Visualizing Adaptation

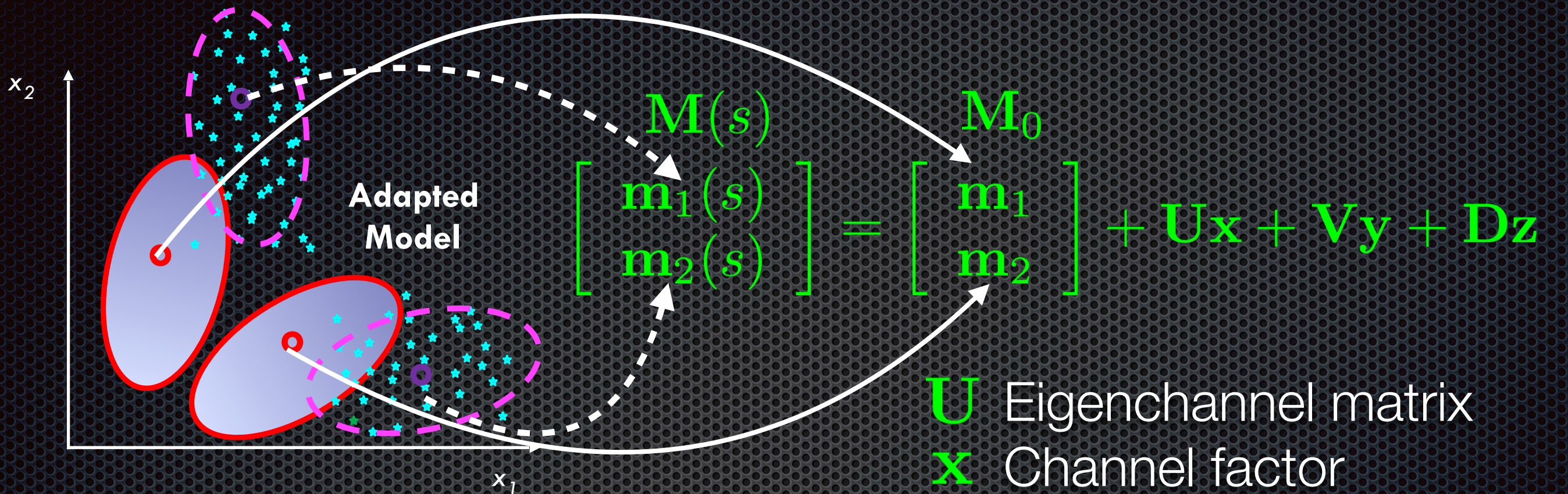


The Millennial BreakThrough 2000-10

- The high fidelity model



Joint Factor Analysis



- Standard Gaussian prior on all factors.
- Model trained using all the training data - Expectation Maximization (EM) algorithm.

i-vector model

$$\mathbf{M}(r) = \mathbf{M}_0 + \mathbf{T}\mathbf{w}(r)$$

$\mathbf{T}$ Total variability matrix
 $\mathbf{w}$ recording factor

- The total variability matrix is trained on the entire training dataset.
- The posterior estimate $\mathbf{w}^*(r) = \operatorname{argmax}_{\mathbf{w}} p(\mathbf{w}|\mathbf{X}(r))$
 - is called the i-vector feature.
- Transformation of the i-vector
 - Length normalization + Linear Discriminant Analysis (LDA).

Dehak, N., Kenny, P. J., Dehak, R., Dumouchel, P., & Ouellet, P. (2011). Front-end factor analysis for speaker verification. *IEEE Transactions on Audio, Speech, and Language Processing*, 19(4), 788-798.

Garcia-Romero, D., & Espy-Wilson, C. Y. (2011). Analysis of i-vector length normalization in speaker recognition systems. In *Interspeech*



Probabilistic Linear Discriminant Analysis (PLDA)

$$\mathbf{w}(r) = \hat{\mathbf{w}} + \mathbf{U}_1 \mathbf{x}_1(s) + \mathbf{U}_2 \mathbf{x}_{2r} + \epsilon_r$$

$\mathbf{U}_1$ Eigenvoice matrix

$\mathbf{x}_1$ Speaker factor

$\mathbf{U}_2$ Eigenchannel matrix

$\mathbf{x}_{2r}$ Recording factor

ϵ_r Residual

- Using a standard Gaussian prior.
- Scoring using PLDA computes

$$\Lambda(\mathbf{X}_e, \mathbf{X}_t) = \log p(\mathbf{w}_e, \mathbf{w}_t | s, r_e, r_t) - \log p(\mathbf{w}_e, \mathbf{w}_t | s_e, s_t, r_e, r_t)$$

- can be efficiently computed with Gaussian model.



Prince, S. J., & Elder, J. H. (2007, October). Probabilistic linear discriminant analysis for inferences about identity. In *IEEE ICCV*.

Kenny (2010), Bayesian speaker verification with heavy tailed priors, *Odyssey*.



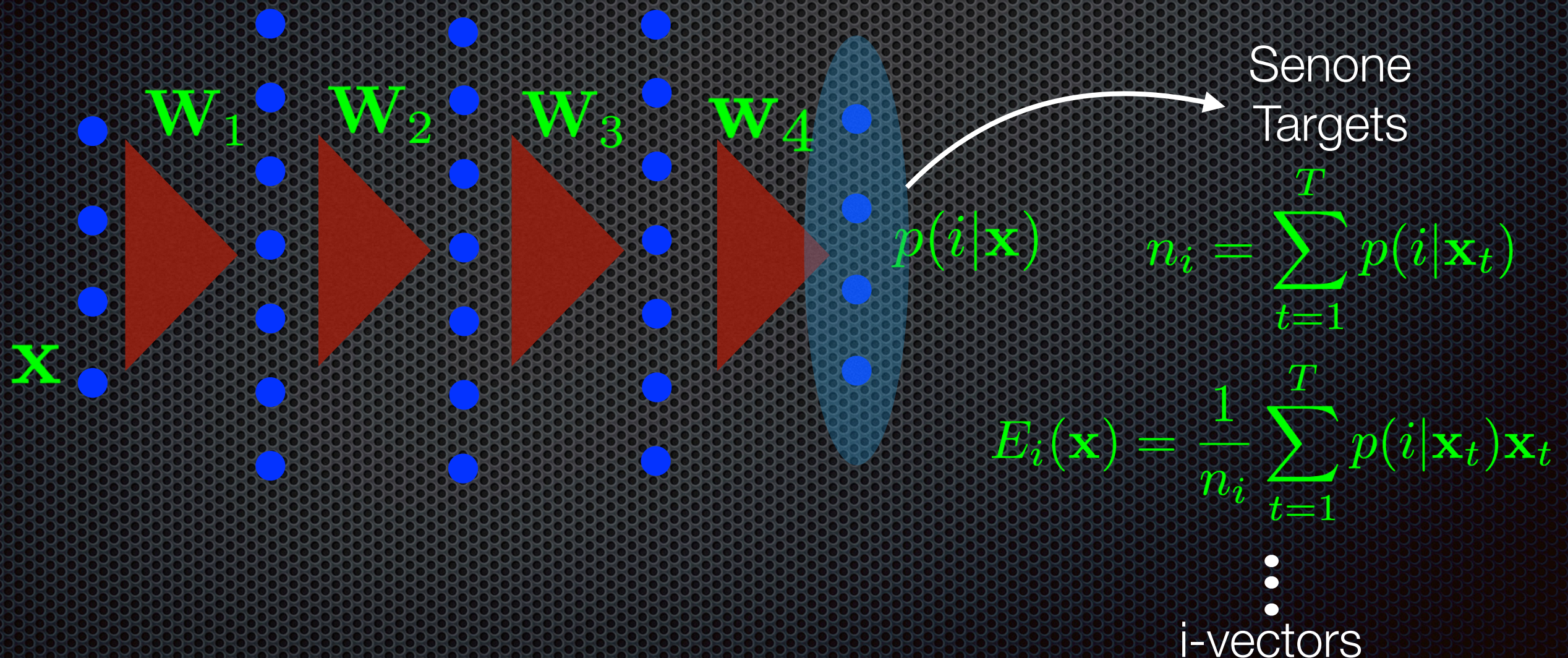
The Deep Revolution > 2011

- The electric version



Speech Recognition Models

- Replacing the background model of Gaussian mixture model (GMM) with deep neural network model trained for speech recognition.



SRE 2010 Results

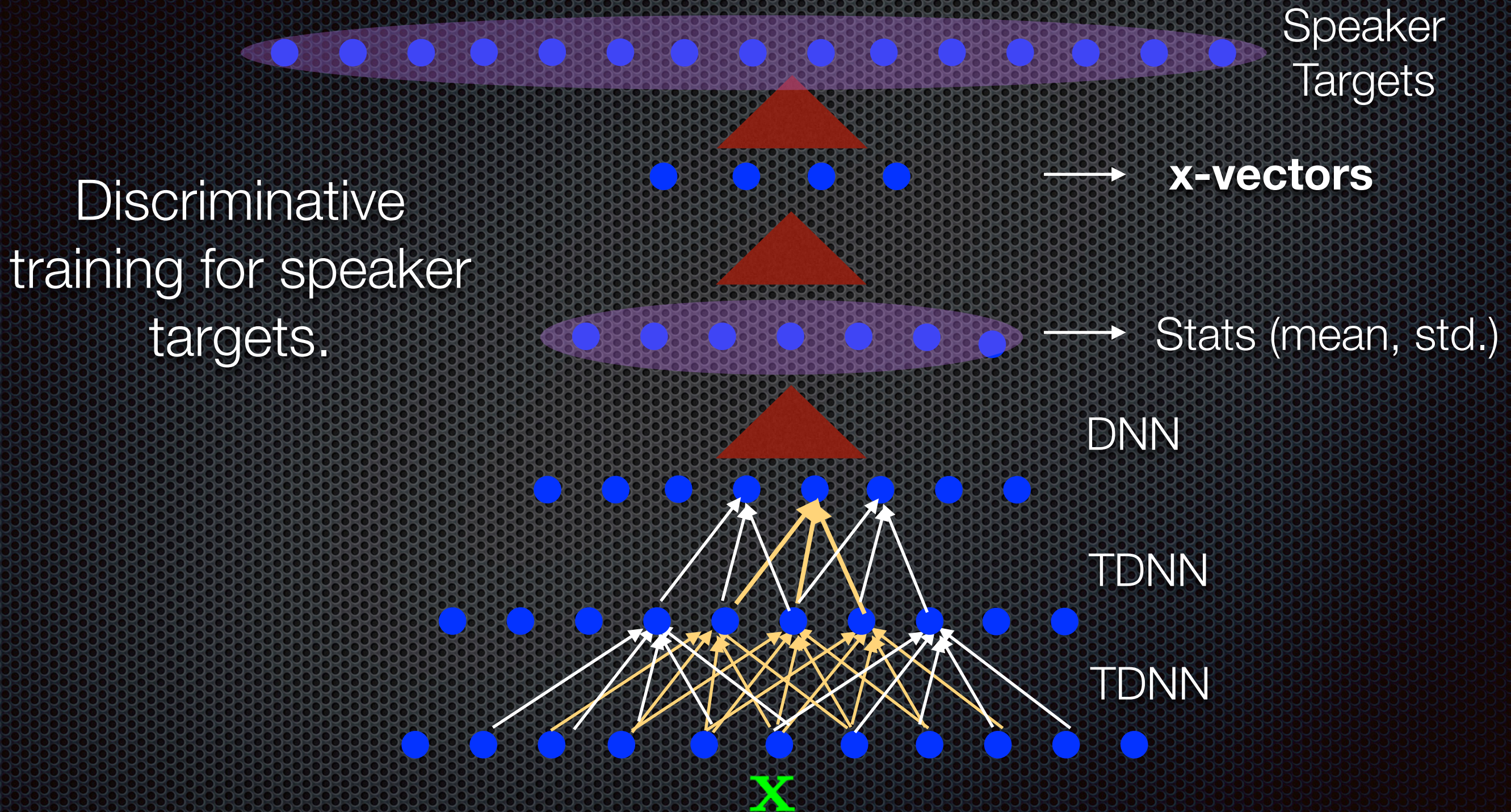
- NIST Speaker Recognition Evaluation 2010 Task.

Model	EER (%)	DCF
GMM-ivector (2048)	2.4	0.44
DNN-ivector (2k Senones)	1.2	0.21
DNN-ivector (4k Senones)	1.0	0.17
DNN-ivector (10k Senones)	0.8	0.12

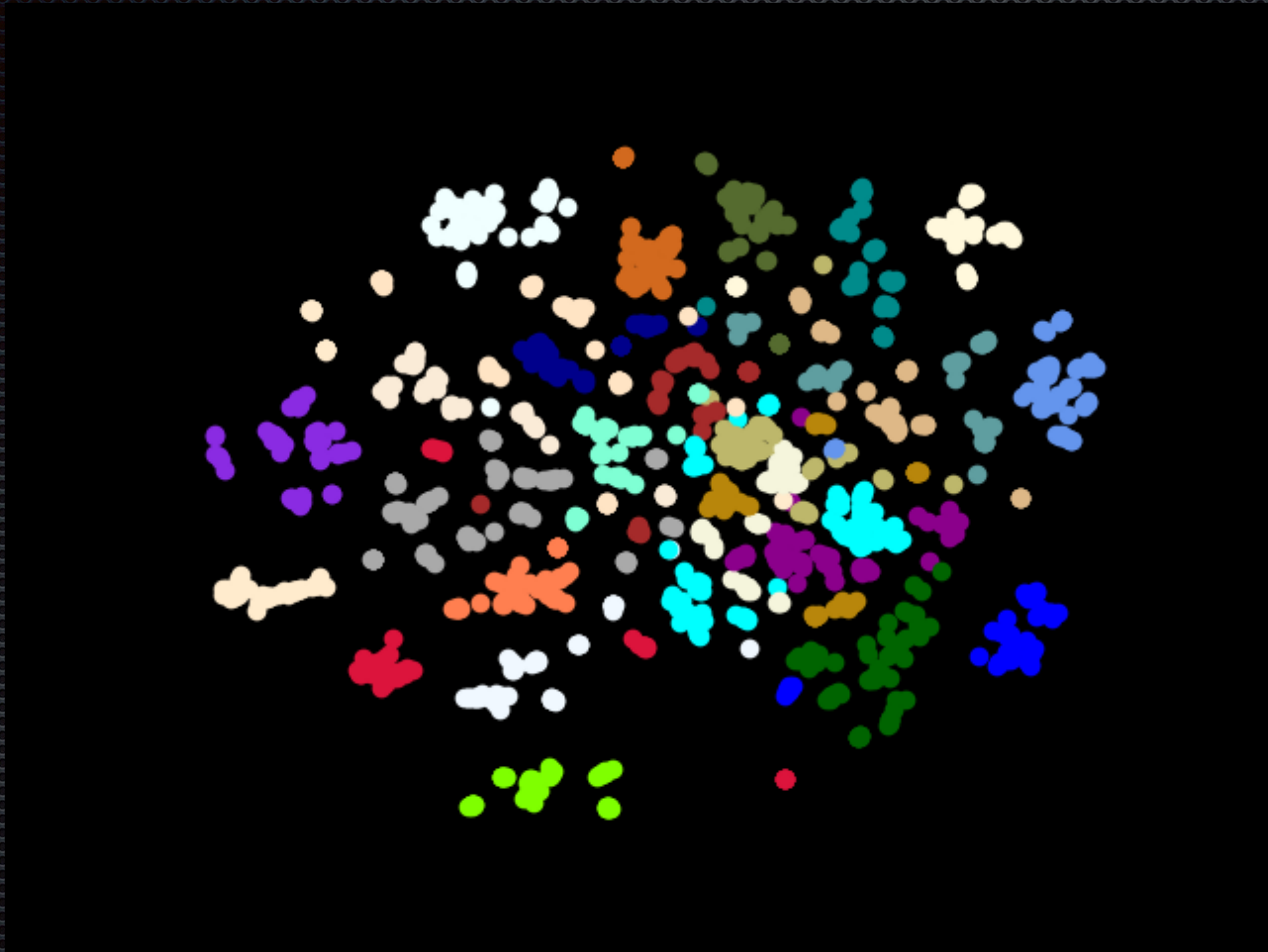
- Best published results in NIST SRE10.



x-vector modeling

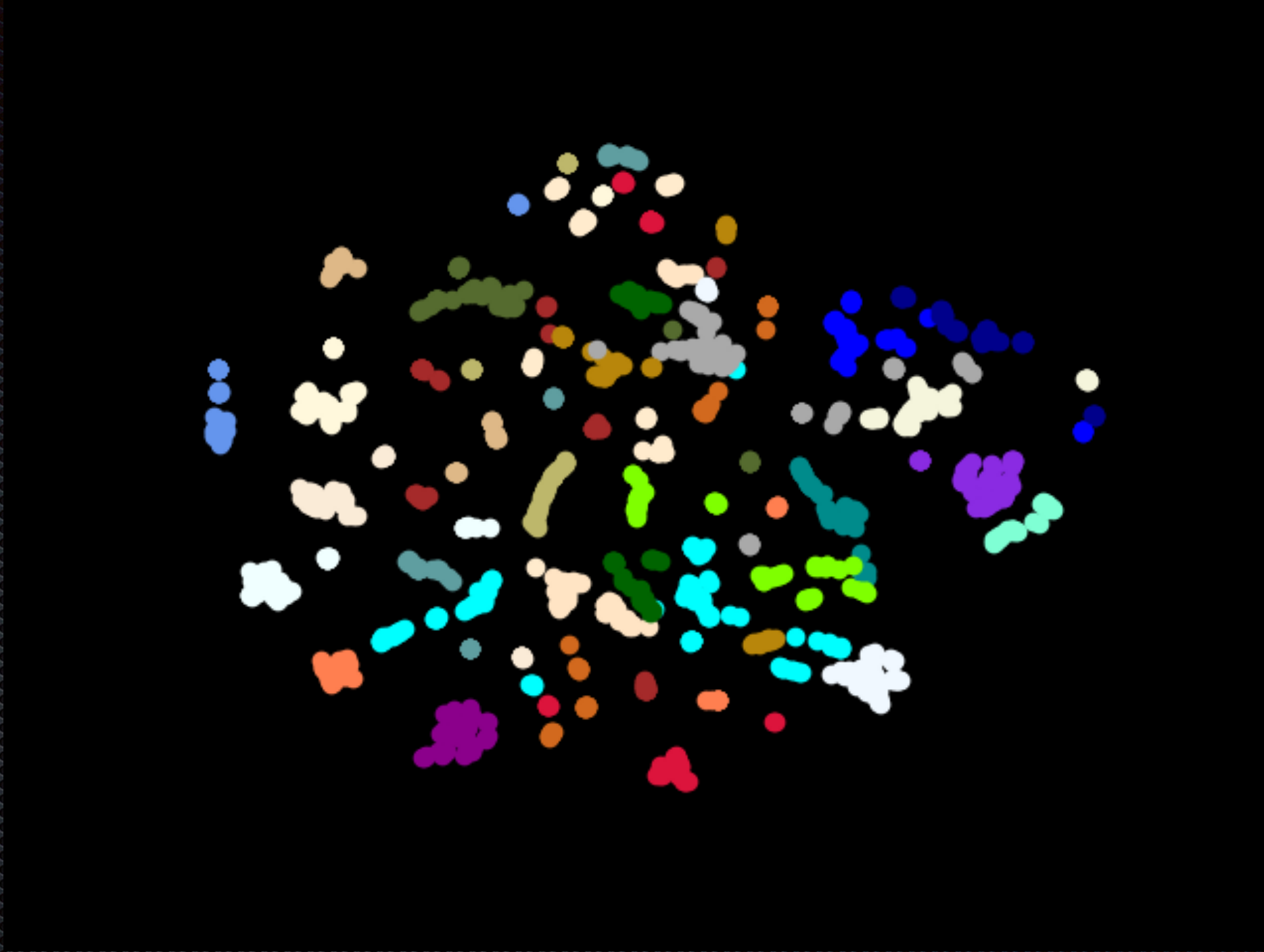


Data visualization (tSNE) i-vector



Colors represent speakers

Data visualization (tSNE) x-vector



Colors represent speakers

SRE 2016 Results

- NIST Speaker Recognition Evaluation 2016 Task.

Model	EER (%)	min Cost
GMM-ivector (2048)	13.1	0.72
DNN-ivector (2k Senones)	8.7	0.61

- Significant improvements with x-vector modeling.



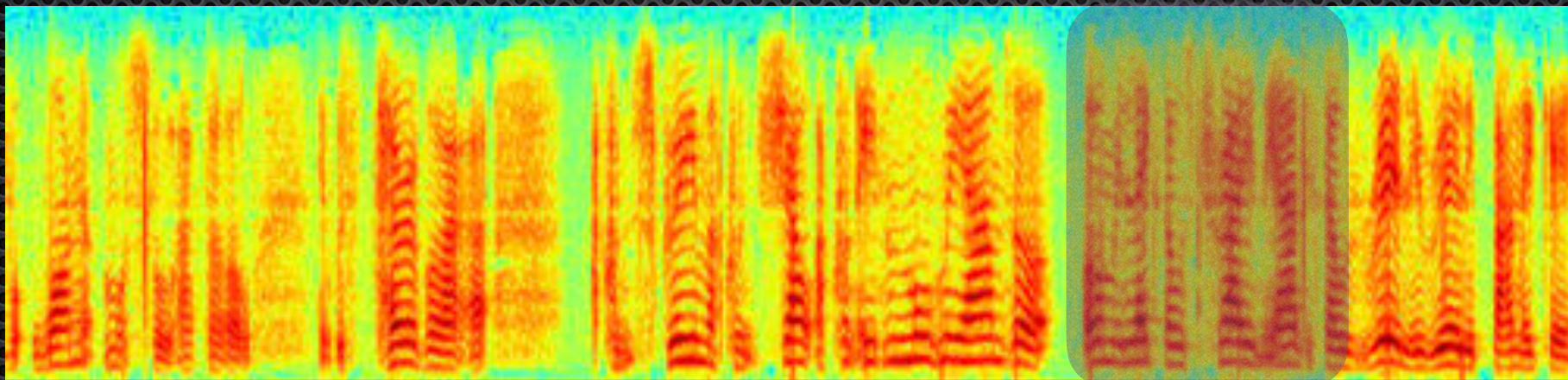
Future Directions

▪ Driverless



Need for Relevance

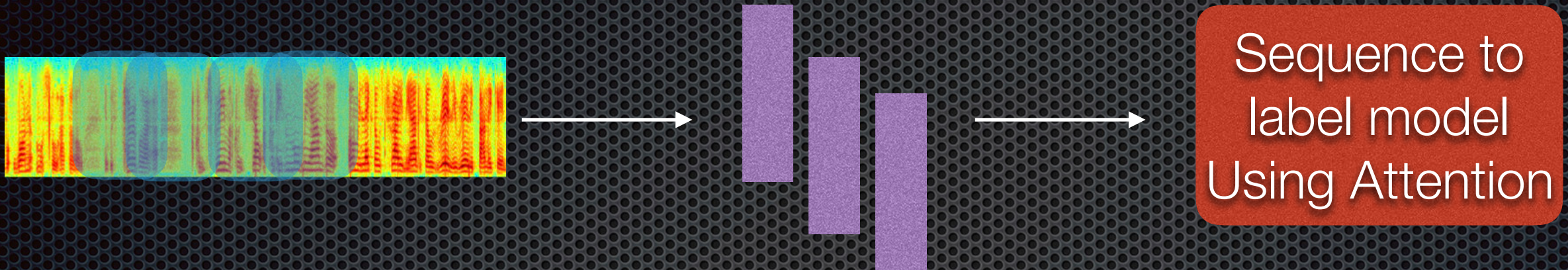
- Certain regions of the audio signal may have more information for the task than the rest.
 - ✓ May also have more signal quality than rest.
 - ✓ For example, language identification involving Eng-UK v/ s Eng. US.



- Current models - use the information from audio with uniformity.

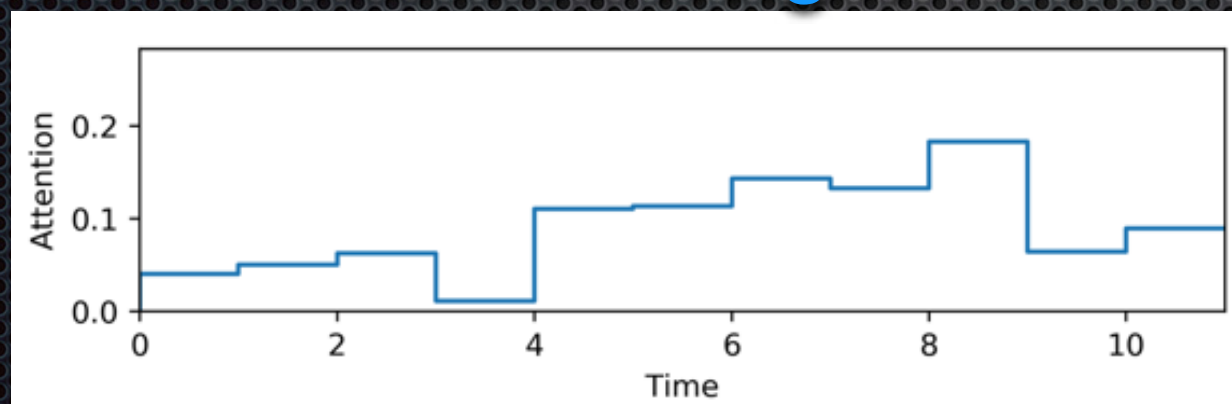
Step in the Direction

- Derive short segment i-vectors



- Attention weighs the importance of each short-term segment feature for the task.

Attention Weight



0-3s : O...One muscle at all, it was terrible

3s-4s : ah ah

4s - 9s : **I couldn't scream, I couldn't shout, I couldn't even move my arms up, or my legs**

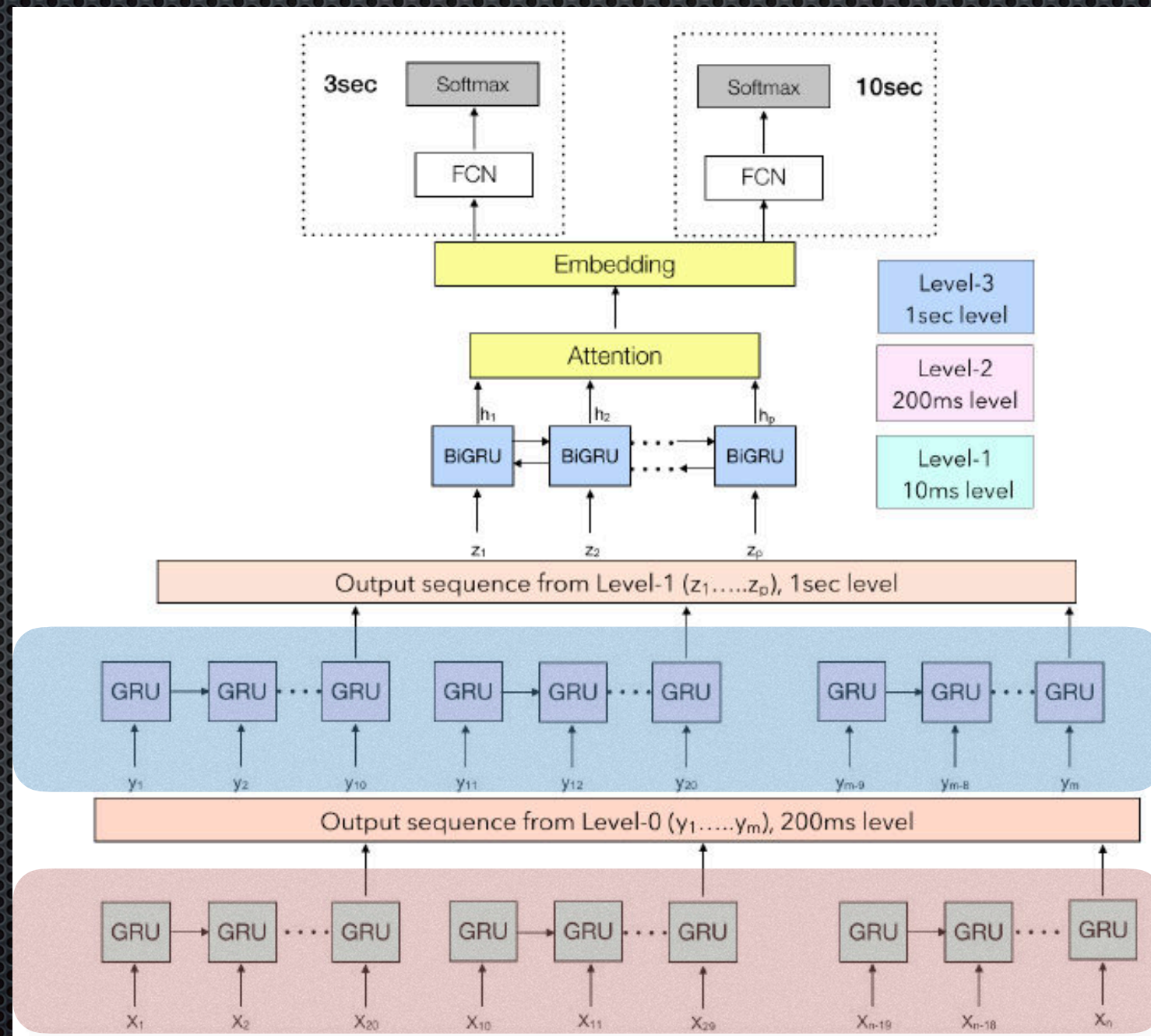
9s -11s : I was trying the attic, I was really really panicking.

End-to-end Models

- Can we model speaker/language end-to-end without individual optimization of blocks.

Step in the direction for language Recognition

Hierarchical Gated Recurrent Unit (HGRU) Modeling.
[Odyssey, 2018]



End-to-end versus Conventional

- NIST Language Recognition Challenge 2017 (Cavg/Acc.).

Model	3s	10s	30s	Multi-talker (30s)
i-vec. + LDA + SVM	0.53 (53.8%)	0.27 (72.4%)	0.13 (83.0%)	0.26 (75.4%)
HGRU + Attn (End-to-end)	0.55 (55.1%)	0.23 (74.1%)	0.18 (83.0%)	0.18 (82.4%)

- Results comparable with state-of-art on all durations
- Significant benefits on multi-talker language recognition.

Challenges & Opportunities

- Multi-task and adversarial learning
 - Dealing with multiple speech cues simultaneously.
- Understanding deep models for speaker and language models
 - Explainable AI.
- Moving to wilder conditions of noisy multi-talker conversational data.

Thanks

